

Learning to Nudge: A Scalable Barrier Function Framework for Safe Robot Interaction in Dense Clutter

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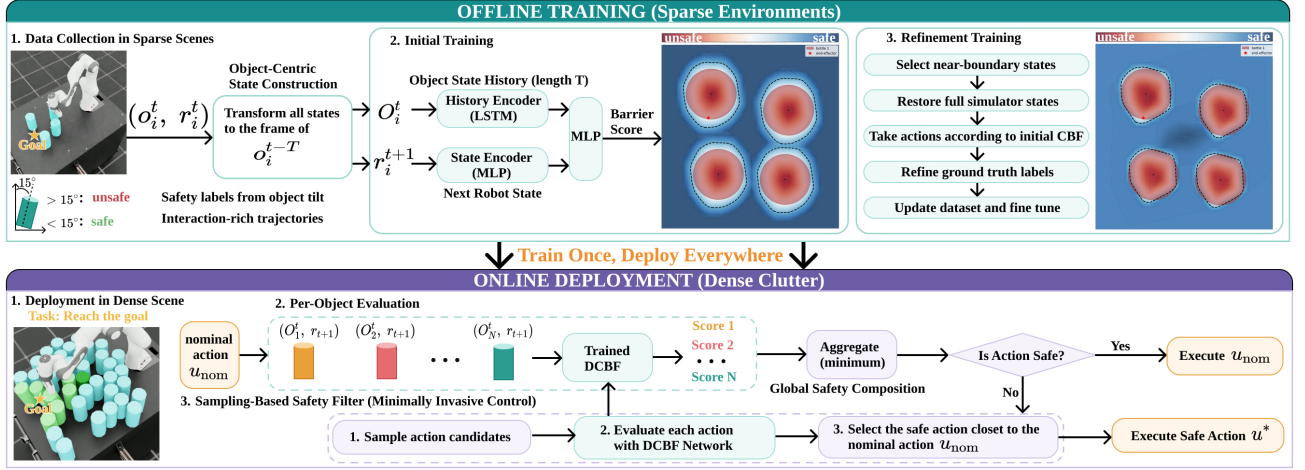


Fig. 1: We train a safety filter specialized for dense clutter only using 4 objects. By redefining safety from collision avoidance to safe interaction, our method enables effective manipulation in dense environments that contain up to 40 objects.

Abstract—Robots operating in everyday environments must navigate and manipulate within densely cluttered spaces, where physical contact with surrounding objects is unavoidable. Traditional safety frameworks treat such contacts as unsafe, which limits their ability to function in dense settings that require gentle contacts. Although model-based approaches to handle such contacts exist, they become computationally intractable as the number of objects increases. Meanwhile, learned methods typically tie safety to the learned task, making it difficult to differentiate and transfer safety from the task itself. To address these problems, we introduce Dense Contact Barrier Functions (DCBF), an approach that bypasses the computational complexity of explicitly modeling multi-object dynamics by instead learning a composable, object-centric function that implicitly captures the safety constraints arising from physical interactions and can be applied on top of any nominal motion policies as a stand-alone safety filter. Trained offline on interactions with only a few objects, the learned DCBF generalizes to settings with significantly more objects without further training, providing a global safety filter that scales linearly with the number of objects. We validate our approach through simulated and real hardware experiments in dense clutter of up to 40 obstacles, demonstrating its ability to enable safe, contact-rich interaction, outperforming baselines by up to 13% overall safe task success.

I. INTRODUCTION

The ability for robots to safely interact with and maneuver through densely cluttered environments is critical for their deployment in everyday, unstructured spaces. These environments often contain tightly packed movable objects, making

physical contact difficult or even impossible to avoid during task execution. For example, a robot retrieving an item from a pantry may need to nudge aside other items, while a logistics robot may need to make contact with fragile goods without knocking them over. Thus, in dense clutter, contact itself is not always a safety violation.

Most existing approaches share a common safety assumption: maintain separation from obstacles, either through safety filtering [1], [2] or collision-free motion generation [3], [4]. This collision-avoidance view also extends to manipulation, where the manipulated object is often incorporated into the set of items with which collision must be avoided [5], [6]. In dense clutter of movable objects, however, strict collision avoidance can be unnecessarily restrictive: a robot that strictly avoids contact may fail to reach otherwise accessible goals when gentle nudging of movable objects is required. Treating every contact as unsafe therefore conflates benign interaction with truly hazardous behavior. A useful safety model for dense clutter should instead determine whether a candidate interaction preserves the stability of the surrounding objects.

Allowing contact, however, makes safety reasoning substantially more challenging, because the outcome of a robot action depends on coupled robot-object and object-object interactions. One approach is to explicitly model these interaction dynamics, which can capture contact-rich behavior but becomes increasingly expensive as the number of potentially

interacting objects grows [7]. Another approach is to learn the behavior directly, for example through reinforcement learning, but such methods typically optimize safety together with task completion, making the learned notion of safety difficult to reuse independently of the task policy. What is therefore missing is a scalable safety representation that can reason about the consequences of contact without explicitly modeling the full multi-object dynamics, while remaining separate from the nominal task controller.

To this end, we introduce Dense Contact Barrier Functions (DCBF), an object-centric control barrier function that learns to estimate stability based on observable object properties. Rather than predicting how an entire scene of N objects may react to the robot’s motion, DCBF expresses the robot’s motion in each object’s frame, together with a window of the object’s own state, to determine whether that object will remain safe. Specifically, an object’s state implicitly captures neighbor interactions, such as tilting when pushed, without requiring explicit multi-object modeling.

We train our DCBF in scenes with only 4 objects, introducing enough interactions to learn contact-induced motion, allowing runtime evaluation to assess each object independently and taking the minimal safety as a global barrier. This composition is linear in the number of objects and transfers to far denser and more varied scenes without retraining. During training of our DCBF, we leverage a two-staged process in which initial barrier violations are refined by re-simulating near-boundary states under the safest available action and relabeling the outcome, which restores contact-rich states that are safe and keeps the filter from stalling in otherwise safe settings. Our primary contributions are:

- a redefinition of safety in dense clutter from collision avoidance to per-object stability, allowing contact while empirically ensuring safety;
- a composable object-centric framework that infers unmodeled inter-object effects;
- a refinement procedure that leverages near-boundary states to recover safe contact-rich states and then reduce the conservativeness of the learned CBF;
- a validation of our approach in simulation and real-world settings across object configuration and physical properties, showing that our DCBF can seamlessly apply as a safety filter on any nominal motion policies.

II. RELATED WORK

In this section, we position our work in the context of safe autonomy, scalability, and control barrier functions, highlighting the limitations and opportunities of current work.

A. Safe Autonomy for Collision Avoidance

Many existing safety approaches primarily focus on avoiding collisions such as Hamilton-Jacobi Reachability (HJR) compute control-invariant safe sets and can be used as minimally invasive safety filters [1], [8]. While learning-based approximations improve scalability [9], [10], these methods rely on structured dynamics models and are primarily demonstrated for collision-avoidance settings.

This collision-avoidance view also appears in manipulation, where potential-field and MPC-based approaches maintain separation or limit contact with surrounding objects [4], [11], [12]. Recent methods extend collision avoidance to cluttered or deformable manipulation by treating the manipulated object as part of the protected system [6], [5]. Although effective when contact can be avoided, such formulations become overly restrictive in dense clutter where interaction with movable objects is often necessary.

B. Challenges in Scalability and Generalization

In settings with more complex interaction dynamics, modeling and generalizing such behaviors becomes increasingly difficult at scale. Learning-based methods such as reinforcement learning integrate safety as a soft constraint [13], [14] or by using Bellman-inspired updates [15], [16], but safety is typically coupled to a particular task or policy, limiting their use as reusable online safety filters. Other approaches leverage advanced simulation and learned dynamics to reason about complex physical interactions. For example, JIGGLE [17] uses XPBD [18] for model-based planning under safety constraints, while [7] learns latent dynamics for safety-aware deformable manipulation and [19] incorporates force feedback for safe articulated-object manipulation. Despite their ability to capture richer interaction effects, these methods are computationally intensive or tailored to limited object settings, making them difficult to scale to variable-sized, multi-object clutter.

C. Control Barrier Functions

A CBF is a function that defines a control invariant safe set in the system’s state space, guaranteeing the existence of a control action that keeps the agent in this set [2], [20]. In online settings, CBFs have been widely used to enforce safety as minimally invasive safety filters, with extensions such as Robust CBFs (RCBFs) [21] generalizing them to uncertain dynamics. Classical CBF synthesis generally requires known system dynamics and carefully designed safety functions [22], while Hamilton–Jacobi-based synthesis scales poorly with state dimension [8]. To overcome this, Neural Control Barrier Functions (NCBFs) learn safety constraints directly from data [23], enabling a new class of methods that incorporate perception [24], handle uncertainty [25], and scale to dense environments [26]. Other works continue to build upon this, enabling refining candidate NCBFs with HJR [27]. Works such as [26] have improved the scalability of safety in dense environments, leveraging composable CBFs to enable safe navigation in crowds. Despite this progress, the application of CBFs remains focused on collision avoidance, leaving a critical gap in methods that can manage safe physical interaction.

In contrast, we learn object-centric barrier functions that characterize the safety of physical interaction rather than contact occurrence itself. Their compositional structure allows the resulting safety filter to operate over a variable number of movable objects in dense clutter.

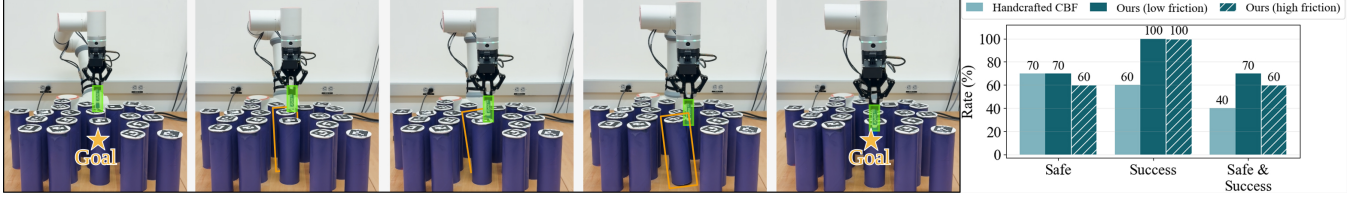


Fig. 2: **Real World Experiment.** Zero-shot real-world evaluation with the Fairino FR5 in 20-object clutter. Left: the robot safely nudges surrounding objects while reaching the goal. Right: performance of RRT paired with different safety filters: a handcrafted CBF, DCBF on a bare table (low friction), and DCBF on a towel-covered table (high friction).

III. PRELIMINARIES

This section introduces the notation and background used in the paper, as well as our general problem statement.

A. Control Barrier Functions

Consider a discrete-time system with dynamics $\mathbf{x}^{t+1} = f(\mathbf{x}^t, u^t)$, where $\mathbf{x} \in \mathbb{R}^n$ and $u \in \mathcal{U}$. A set $\mathcal{C} \subseteq \mathbb{R}^n$ is said to be *control-invariant* if, for every $\mathbf{x} \in \mathcal{C}$, there exists an admissible control input $u \in \mathcal{U}$ such that

$$f(\mathbf{x}, u) \in \mathcal{C}.$$

Given a function $h : \mathbb{R}^n \rightarrow \mathbb{R}$, define its superlevel set as

$$\mathcal{C}_h := \{\mathbf{x} \in \mathbb{R}^n \mid h(\mathbf{x}) \geq 0\}.$$

We say that h is a discrete-time *control barrier function* (CBF) if, for every $\mathbf{x} \in \mathcal{C}_h$, there exists an admissible control input $u \in \mathcal{U}$ such that

$$h(f(\mathbf{x}, u)) - h(\mathbf{x}) \geq -\alpha(h(\mathbf{x})), \quad (1)$$

where $\alpha(\cdot)$ is an extended class- $\mathcal{K}$ function satisfying

$$0 \leq \alpha(r) \leq r, \quad r \geq 0. \quad (2)$$

Under this condition,

$$h(f(\mathbf{x}, u)) \geq h(\mathbf{x}) - \alpha(h(\mathbf{x})) \geq 0,$$

for every $\mathbf{x} \in \mathcal{C}_h$. Therefore, the superlevel set $\mathcal{C}_h$ is control-invariant.

A commonly used special case is the linear choice

$$\alpha(r) = \gamma r, \quad \gamma \in (0, 1], \quad (3)$$

for which (1) becomes

$$h(\mathbf{x}^{t+1}) \geq (1 - \gamma)h(\mathbf{x}^t). \quad (4)$$

Since $1 - \gamma \geq 0$, an initial state satisfying $h(\mathbf{x}^0) \geq 0$ remains in $\mathcal{C}_h$ for all subsequent time steps.

B. Problem Statement

We consider the problem of a robot safely moving through an environment that is densely populated with obstacles. In this setting, the robot may physically interact with the objects, but must ensure that its actions do not cause the objects to enter an unsafe state, such as tilting too far or toppling over. This setting is relevant to a variety of manipulation tasks such as reaching for a glass in a pantry, a ground robot pushing aside furniture, or a warehouse robot reorganizing shelves. In each case, safety is defined in terms

of object stability, while the robot must remain effective in task execution.

To formulate this problem, we define the state of the robot as $\mathbf{r} \in \mathbb{R}^n$ with dynamics

$$\mathbf{r}^{t+1} = f(\mathbf{r}^t, u^t), \quad (5)$$

where $u^t \in \mathbb{R}^a$ is the control input. The environment contains N objects, each with a state $\mathbf{o}_i \in \mathbb{R}^m \forall i \in [1, N]$. The safety of each object is defined by a function $q_i(\mathbf{o}_i)$, where a violation occurs if this safety cost exceeds a known threshold, i.e. $q_i(\mathbf{o}_i) \geq q_T$.

For a scene with a single object, $\mathbf{o}$, the dynamics is given by: $\mathbf{o}^{(t+1)} = p(\mathbf{r}^t, u^t, \mathbf{o}^{(t)})$. However, for multiple objects, the dynamics, given by

$$\begin{aligned} &\{\mathbf{o}_1^{(t+1)}, \mathbf{o}_2^{(t+1)}, \dots, \mathbf{o}_N^{(t+1)}\} \\ &= F(\mathbf{r}^{(t)}, u^{(t)}, \{\mathbf{o}_1^{(t)}, \mathbf{o}_2^{(t)}, \dots, \mathbf{o}_N^{(t)}\}), \end{aligned} \quad (6)$$

are influenced by both the robot and inter-object interactions. A key challenge with reasoning over the interactions in F is that the number of inter-object interactions scales exponentially with the number of objects. Reasoning about the safety of each object by explicitly modeling it becomes computationally infeasible, especially in online settings. We assume access to the robot's forward dynamics (5), but no runtime access to the object dynamics (6). The problem then becomes to define a safety filter that, given a nominal control action u_{nom} , returns an admissible control keeping every object within its safety threshold, $q_i(\mathbf{o}_i) < q_T$ for all i , while deviating from u_{nom} as little as possible. Such a filter must function without F , its cost should scale favorably with the number of objects N , and should generalize to various values and arrangements of N without requiring further training.

IV. DENSE CONTACT BARRIER FUNCTIONS

Fig. 1 provides an overview of the proposed training and deployment pipeline. We aim to implicitly capture inter-object contact dynamics using a data-driven neural CBF that estimates safety on a per-object basis. The proposed model is conditioned on the robot's state expressed in an object-centric frame and on the object's interaction history. At inference, these per-object barrier values are composed into a global safety assessment, resulting in a safety filter that is separate from the nominal task controller and permits safe contact while generalizing to unseen scene densities. We start by showing how a bounded model error is combined with a robustness margin σ , then argue that learning the object

transition explicitly is impractical in contact-rich settings, and finally show that the same margin lets us impose the barrier condition directly on a learned safety landscape.

A. Implicit Interaction

Explicitly modeling object dynamics in contact-rich clutter is challenging, as object motion depends on contact geometry, friction, physical properties, and interactions with surrounding objects. Instead of predicting these dynamics, we directly learn a barrier function that captures their safety-relevant effects from observed motion. We first consider a single-object setting with known robot dynamics (5) and unknown contact-driven object dynamics $\mathbf{o}^{t+1} = p(\mathbf{o}^t, \mathbf{r}^t, u^t)$. Following Sec. III-A, we use the linear discrete-time CBF choice $\alpha(s) = \gamma s$, with $\gamma \in (0, 1]$, throughout the remainder of the paper. Given a discrete-time barrier function $h(\mathbf{r}^t, \mathbf{o}^t)$, safety under the true dynamics requires

$$h(\mathbf{r}^{t+1}, \mathbf{o}^{t+1}) - h(\mathbf{r}^t, \mathbf{o}^t) \geq -\gamma h(\mathbf{r}^t, \mathbf{o}^t). \quad (7)$$

However, evaluating this condition directly requires the contact dynamics p , which are difficult to model accurately due to nonlinear, non-smooth, and environment-dependent interactions.

Rather than explicitly learning p , we learn a safety predictor $B(\mathbf{r}^{t+1}, \mathbf{o}^t)$ that directly estimates the safety consequence of a candidate action:

$$B(\mathbf{r}^{t+1}, \mathbf{o}^t) \approx h(\mathbf{r}^{t+1}, \mathbf{o}^{t+1}). \quad (8)$$

This allows the effect of object interaction to be captured implicitly from data while requiring only the known current object state $\mathbf{o}^t$ together with $\mathbf{r}^{t+1}$, which can be computed from the known robot dynamics.

Because B is an approximation, we introduce a robustness margin $\sigma > 0$. Suppose its overestimation of the true next-step barrier value is bounded as

$$B(\mathbf{r}^{t+1}, \mathbf{o}^t) - h(\mathbf{r}^{t+1}, \mathbf{o}^{t+1}) \leq \sigma. \quad (9)$$

Then enforcing

$$B(\mathbf{r}^{t+1}, \mathbf{o}^t) - h(\mathbf{r}^t, \mathbf{o}^t) - \sigma \geq -\gamma h(\mathbf{r}^t, \mathbf{o}^t) \quad (10)$$

is sufficient to satisfy the true DCBF condition in (7). Thus, σ provides a robustness margin against errors introduced by implicitly predicting the effect of contact.

In the following section, we use this principle to learn the discrete-time barrier function directly from interaction data, with σ controlling the conservativeness required to account for prediction uncertainty.

B. Barrier Formulation

1) *Temporal Object-State History*: Following [26], we provide the barrier with a finite window of object states rather than a single observation. We define an absolute history window of length T as $\mathbf{O}^t = [\mathbf{o}^t, \mathbf{o}^{t-1}, \dots, \mathbf{o}^{t-(T-1)}]$, which B encodes with a recurrent module.

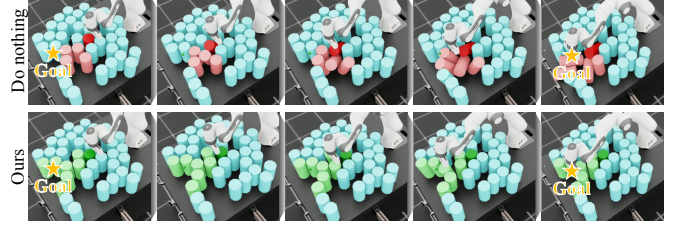


Fig. 3: **Example rollouts** of Do Nothing and Do Nothing + DCBF in dense clutter. The unfiltered controller destabilizes surrounding objects, whereas DCBF filters unsafe actions and reaches the goal while maintaining object stability.

2) *Object-Centric Representation*: Directly learning a joint barrier over the full multi-object state is generally intractable due to the combinatorial growth of interaction modes. Inspired by [28], we adopt an object-centric representation to learn a single per-object model that can be deployed across variable-sized object sets. Let $R(a, b)$ denote a transformation that expresses state a relative to reference b . For each object i , we anchor the coordinate frame at the object’s position at time $t - T$ and define

$$\mathbf{r}_i^t = R(\mathbf{r}^t, \mathbf{o}_i^{t-T}),$$

$$\mathbf{O}_i^t = [R(\mathbf{o}_i^t, \mathbf{o}_i^{t-T}), \dots, R(\mathbf{o}_i^{t-(T-1)}, \mathbf{o}_i^{t-T})].$$

In this representation, deviations from the “expected” object motion (e.g., lateral drift when pushed forward) implicitly encode the presence of object-object interactions.

3) *Inference for Multi-Object Scenes*: To assess scene-level safety, we compose the per-object barrier values via a minimum operator. For N objects, the safety for a candidate control input u is evaluated as:

$$B_{\text{global}}^u(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) = \min_{i=1, \dots, N} B(f(\mathbf{r}_i^t, u), \mathbf{O}_i^t). \quad (11)$$

A control input is treated as admissible if $B_{\text{global}} \geq 0$, which is equivalent to requiring that all object-level barrier values are nonnegative (safe). This composition scales linearly in N and is trivially parallelizable across objects.

C. Control Barrier Training and Deployment

In this section, we describe the training procedure for the proposed DCBF and show how it is deployed online as a minimally invasive, sampling-based safety filter. The training process involves three stages: 1) **interaction-rich data collection**, 2) **initial training** of a neural barrier model, and 3) **iterative boundary refinement** to reduce conservativeness and improve invariance. Online execution then uses the refined model to filter a nominal controller.

1) *Data Collection*: We collect interaction-rich trajectories in simulation using a suboptimal behavior policy. This policy induces both safe and unsafe behaviors, yielding frequent interactions near the safety boundary without requiring a highly tuned controller. Training scenes contain only a finite number of objects, typically far fewer than at deployment.

For each trajectory and each object i , we store transition tuples $\{(\mathbf{r}_i^t, \mathbf{O}_i^{t-1}), (\mathbf{r}_i^{t+1}, \mathbf{O}_i^t)\}$, and assign a binary label using the ground-truth safety criterion at the resulting physical state. The labeled transitions form a safe dataset $\mathcal{D}_s$ and an unsafe dataset $\mathcal{D}_u$, with $\mathcal{D} := \mathcal{D}_s \cup \mathcal{D}_u$. For later refinement, we also store the full simulator state at each transition to enable re-simulation under alternative control inputs.

2) *Initial Training*: We parameterize B with a neural network comprising (i) an LSTM encoder for the temporal object history, (ii) an MLP encoder for the robot state, and (iii) a fusion head that outputs a scalar barrier value. We obtain an initial model B_θ by minimizing the composite objective $L(\theta)$ defined as:

$$L(\theta) = \eta_s L_s(\theta) + \eta_u L_u(\theta) + \eta_d L_d(\theta), \quad (12)$$

$$L_s(\theta) = \mathbb{E}_{\mathcal{D}_s} \left[-B_\theta(\mathbf{r}_i^t, \mathbf{O}_i^{t-1}) \right]_+, \quad (13)$$

$$L_u(\theta) = \mathbb{E}_{\mathcal{D}_u} \left[B_\theta(\mathbf{r}_i^t, \mathbf{O}_i^{t-1}) \right]_+, \quad (14)$$

$$L_d(\theta) = \mathbb{E}_{\mathcal{D}_s \cup \mathcal{D}_u} \left[\begin{aligned} &(1 - \gamma) B_\theta(\mathbf{r}_i^t, \mathbf{O}_i^{t-1}) \\ &- B_\theta(\mathbf{r}_i^{t+1}, \mathbf{O}_i^t) + \sigma \end{aligned} \right]_+, \quad (15)$$

where $[\cdot]_+$ denotes ReLU, $\eta_s, \eta_u, \eta_d > 0$ are weighting coefficients, and $\gamma \in (0, 1]$ is the scalar coefficient of the linear class- $\mathcal{K}$ choice $\alpha(s) = \gamma s$. The terms L_s and L_u enforce the correct sign on the labeled samples, while L_d penalizes violations of the discrete-time barrier condition.

3) *Refinement*: The initial model, trained on trajectories generated by a suboptimal behavior policy, is typically conservative. Because the labels in $\mathcal{D}$ reflect the outcomes of the behavior policy, many feasible contact-rich states that admit safe actions are nevertheless observed (and labeled) unsafe, introducing false negatives and shrinking the learned safe set. Meanwhile, poor initial training can produce false safe classifications that violate invariance. We address both issues with iterative refinement near the learned boundary.

To identify near-boundary samples, we select transitions from $\mathcal{D}$ such that $|B(\mathbf{r}_i^t, \mathbf{O}_i^{t-1})| \leq \delta$ for a tunable threshold $\delta > 0$. For each selected transition, we re-initialize the full multi-object simulator state and execute a short rollout (up to s steps) under a “safest-action” policy that maximizes the next-step global barrier value:

$$u_{\text{refine}}(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) = \arg \max_{u \in \mathbb{R}^a} B_{\text{global}}^u(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) \quad (16)$$

All transitions collected during these rollouts are annotated using the ground-truth safety criterion and merged into the training set, after which we fine-tune the model via (12). We repeat this loop until performance stabilizes.

At runtime, the refined barrier is used as a safety filter that minimally perturbs a nominal control input u_{nom} to ensure safety. As illustrated in Fig. 4, we discretize the local control space around u_{nom} (Fig. 4 Right) into a finite set of candidate actions and then perform action selection as follows:

- 1) Evaluate the nominal control. If $B_{\text{global}}^{u_{\text{nom}}}(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) \geq 0$, execute u_{nom} .

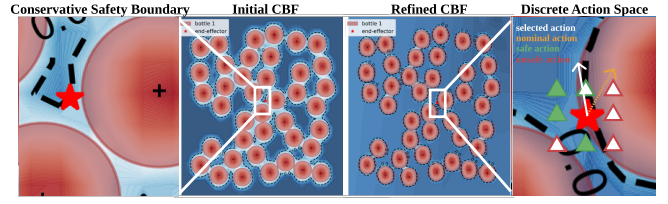


Fig. 4: **Illustration of barrier refinement and action filtering.** The initial CBF produces a conservative safety boundary, which is refined using near-boundary interaction data. At deployment, candidate actions around the nominal action are evaluated by the refined DCBF, and the closest safe action is selected. The asterisk denotes the end effector and the circles denote surrounding objects.

- 2) Otherwise, sample a finite candidate set $\mathbb{U}$ by discretizing the local control space around u_{nom} .
- 3) Form the safe subset $\mathbb{U}_s = \{u \in \mathbb{U} \mid B_{\text{global}}^u(\mathbf{r}_{1:N}^t, \mathbf{O}_{1:N}^t) \geq 0\}$ and execute $u_{\text{filtered}} = \arg \min_{u \in \mathbb{U}_s} \|u - u_{\text{nom}}\|$. If no sampled candidate is admissible, i.e. $\mathbb{U}_s = \emptyset$, the robot stops.

Evaluating B for all objects at every time step can be costly. In practice, it is often sufficient to only evaluate objects that are (i) within a fixed spatial neighborhood of the robot or (ii) exhibit recent motion within their history window, as they are likely to be affected by direct or indirect interactions.

Together, the trained barrier and the sampling filter form a safety layer that requires no task information and no model of the objects at runtime, and whose cost grows linearly in the number of objects evaluated. The same trained model can therefore be placed on top of any nominal controller, which we exploit in the following section by pairing it with several planners of differing sophistication.

V. EXPERIMENTS

We validate our method in both simulation and a real-world scenario across multiple obstacles where physical contact is unavoidable. In both settings, the robot navigates through dense clutter composed of numerous cylindrical objects to reach a target, demonstrating safe and scalable contact-rich interaction while indicating strong generalization from sparse training scenes to substantially denser clutter at deployment.

Experiment Setup: Experiments are conducted using a Franka Emika Panda robot in the Isaac Lab simulator (Fig. 3) and a Fairino FR5 arm in the real world (Fig. 2). The robot must reach a randomly sampled planar target while ensuring that no bottle exceeds a tilt angle of 15° . The end-effector is constrained to planar (x, y) motion, and the action space consists of 2D waypoint increments with a maximum step size of 1cm. Each object state is $\mathbf{o}^t = (x^t, y^t, z^t, \theta^t)$, where (x^t, y^t, z^t) denotes the bottle’s Cartesian position, and θ^t is the tilting angle. Relative transforms are computed by subtracting (x, y) coordinates in the horizontal plane.

To assess robustness to object heterogeneity, we randomize physical parameters across trials, including mass (1.3–2.0kg)

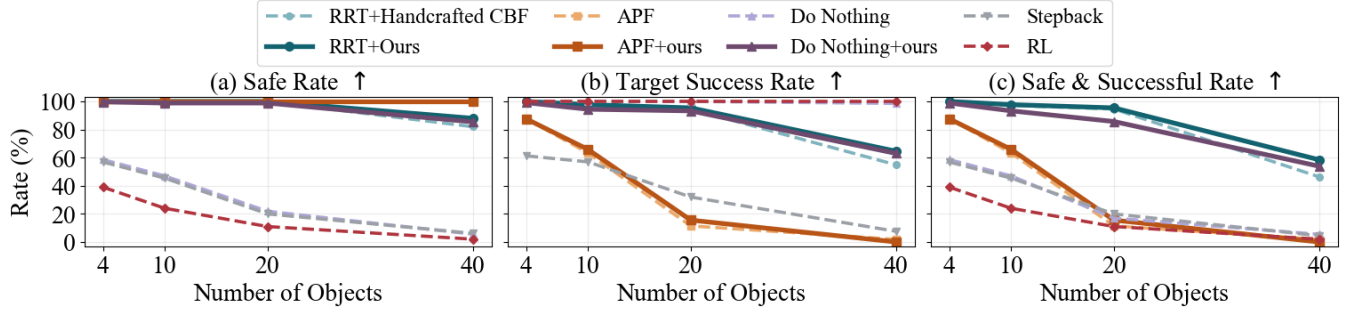


Fig. 5: **Simulation performance** across 100 environments with 7 random seeds from 4 to 40 objects. We report (a) Safe Rate, (b) Target Success Rate, and (c) Safe-and-Successful Rate for nominal controllers with and without our DCBF.

and friction coefficients (static: 0.5–0.7; dynamic: 0.3–0.49). These parameters are not provided as explicit inputs to the learned CBF; the model must therefore remain effective across the randomized range without observing the underlying properties.

Metrics: In all of our experiments, we report Safe Rate, Target Success Rate, and Safe-and-Success Rate. The safe rate captures the percentage of interactions without a safety violation, in this case, when objects are tilted beyond 15° or toppled over, irrespective of overall task success. The target success rate captures the model’s ability to reach the goal irrespective of safety violations. The safe and successful rate is the combination of both metrics.

Implemented details: Models are trained in environments containing only 4 bottles. The initial dataset has 1,200 trajectories collected with a naive back-stepping policy: the robot moves toward a sampled goal, and when any object’s tilt approaches the safety threshold, it backtracks to the previous waypoint, resampling a new goal. To increase interaction richness, we place the objects close to the end effector.

Our model consists of an LSTM with hidden size 64 to encode the history sequence. The final hidden state is concatenated with the planar end-effector position and passed through a two-layer MLP with hidden size 64, which outputs a scalar CBF value. We use a history length of $H = 3$ and train for 60 epochs with batch size 256 and learning rate 10^{-4} , applying weight decay $\lambda = 0.01$. We set the CBF coefficient to $\gamma = 0.1$. Based on the ablation in Sec. V-C, we use robustness margins $\sigma_{\text{init}} = 0.01$ and $\sigma_{\text{refine}} = 0.02$ for training and refinement, respectively. During refinement, each selected near-boundary state is re-simulated for $s = 4$ control steps. Training our model takes approximately 16 hours on an Nvidia H200 GPU.

A. Comparison to Baselines in Simulation

We first evaluate the proposed models in simulation and assess generalization by deploying it in scenes with 4, 10, 20, and 40 heterogeneous objects. We further evaluate our method against the best-performing baseline in a 40-object scenario. For each scenario, we use 7 random seeds, with 100 parallel environments per seed.

We organize the comparisons into three groups: nominal policies, standalone baselines, and safety filters.

Nominal policies. (i) *Do Nothing*, which moves directly toward the goal in a straight line; (ii) *Artificial Potential Fields (APF)* [4], a repulsive-field method with parameters well-tuned to allow limited contact without excessive conservativeness; and (iii) *Rapidly-exploring Random Tree (RRT)* [3], a classical sampling-based planner that generates a collision-avoiding path toward the fixed goal.

Standalone baselines. (i) *Back-stepping*, the reactive policy used for data collection, which retreats the robot to its previous state when the tilt of any nearby object approaches 14° ; and (ii) *RL-Safety*, a reinforcement-learning policy trained end-to-end with a task reward encouraging progress toward the goal and a safety reward penalizing unsafe object states;

Safety filters. To compare against a conventional model-based safety filter, we evaluate *RRT + Handcrafted CBF*. For each obstacle i , we define a distance-based barrier

$$h_i(\mathbf{x}) = \|\mathbf{p}_{\text{robot}} - \mathbf{p}_{\text{obs},i}\| - (r_{\text{robot}} + r_{\text{obs},i}). \quad (17)$$

The nominal control is filtered by solving a QP that minimizes $\|u - u_{\text{nom}}\|_2^2$ subject to $\nabla h_i(\mathbf{x})^\top u + \alpha h_i(\mathbf{x}) \geq 0$ for all obstacles and $\|u\| \leq u_{\text{max}}$.

We then apply our trained DCBF safety filter to the same nominal planners, yielding *Do Nothing + Ours*, *APF + Ours*, and *RRT + Ours*. This comparison isolates the contribution of the learned safety filter and shows how it changes the safety and task performance of different underlying planners. Fig. 4 provides a qualitative visualization induced by the proposed method. The initial model is overly conservative and produces large “blocked” regions that cause the robot to stall (left of Fig. 4). After refinement, the boundary is tighter and admits safe contact-rich pathways that enable goal reaching (right of Fig. 4).

Quantitatively, Fig. 5 reveals three main observations: (i) Collision-avoidance-oriented APF maintains a relatively high safe rate but becomes increasingly conservative as clutter density increases, leading to degraded target success while direct or learned task policies, such as *Do Nothing* and *RL-Safety*, retain relatively high task-completion rates but increasingly violate the object-stability constraint; (ii) Adding DCBF consistently improves the safety–performance trade-off of the nominal controllers, with a particularly pronounced

TABLE I: Safe-and-success rate(%) for different robustness margins during initial training (σ_{init}) and refinement (σ_{refine}) using Do Nothing + DCBF in 40-object scenario.

σ_{init}	σ_{refine}			
	0.01	0.02	0.04	0.05
0.01	9	54	49	50
0.02	6	47	47	36
0.04	3	8	15	17

improvement for Do Nothing and RRT, where DCBF permits contact that is predicted to be safe while rejecting actions likely to induce tipping. Adding DCBF to APF produces relatively small changes, as APF is already highly conservative and therefore rarely proposes contact-rich actions. This pattern is itself evidence that the barrier responds to interaction rather than proximity as the filter contributes in proportion to how frequently the nominal policy proposes contact, and contributes least for the policy that avoids contact by construction. Finally, (iii) these improvements persist as the number of objects increases from 4 to 40, despite the DCBF being trained exclusively in four-object scenes. This result demonstrates that the proposed object-centric composition enables generalization to substantially denser environments without retraining and that DCBF can be applied as a safety-only approach on top of existing motion planners.

B. Evaluation on Real Hardware

We evaluate the proposed model zero-shot on hardware with 20 obstacles across two different friction surfaces after training entirely in simulation. We employ a Fairino FR5 arm and two Orbbec Femto Mega cameras positioned to track the position and tilt-angle of the obstacles. Object poses are tracked using 3D printed ArUco markers mounted on top of the objects. Each object has a radius of 7cm and a height of 23cm with a relatively light weight of approximately 62g, arranged to yield an object density comparable to the 40-object simulation setting. In real-world trials, we define safety as the absence of knocked-over objects rather than enforcing a strict tilt-angle threshold. Table II reports the total computation time per rollout and the mean end-effector jerk on the rigid table and towel-covered surfaces. The computation time remains comparable across the two surfaces (45.51 s on the table and 43.06 s on the towel), indicating that the change in contact conditions does not introduce substantial additional computational overhead. The corresponding mean jerk values are 1.44 and 1.74 mm/s³, respectively, suggesting that the filtered motions remain smooth in both settings. Figure 2 (Right) reports the hardware results over 10 trials each in the 20-object settings on the table surface and high-friction towel. The observed performance mirrors the trends in simulation. The safe rates among three scenarios are similar, but our method has higher success rate on both of the surfaces. Consequently, the filter admits contact-rich actions that let the end effector pass through gaps with slight nudging without compromising safety and retaining task success.

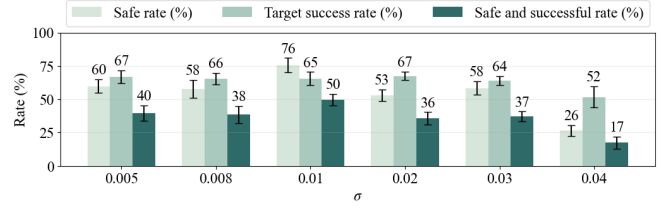


Fig. 6: **Ablation study.** Effect of varying σ_{init} with fixed $\sigma_{\text{refine}} = 0.05$ on the Safe-and-Success Rate (%) for Do Nothing + DCBF in the 40-object scenario.

C. Ablations and Analysis

In this section, we assess our method’s complexity and sensitivity to its safety parameterization. We use the Do-Nothing baseline with our safety filter for this ablation to avoid the substantially higher computational cost introduced by RRT with our safety filter.

1) *Computational Efficiency:* Table II reports the computational efficiency of DCBF as scene density increases. As the number of objects increases from 4 to 40, the computational cost increases predictably with scene complexity, consistent with the object-wise evaluation required by our compositional barrier formulation.

2) *Robustness Margin Ablation:* We first evaluate the effect of varying σ_{init} while fixing $\sigma_{\text{refine}} = 0.05$. As shown in Fig. 6, $\sigma_{\text{init}} = 0.01$ achieves the best Safe-and-Success Rate among the tested values in the 40-object setting. We then investigate the sensitivity of the proposed method to the robustness margin σ during the two stages of training: initial barrier learning and boundary refinement. Specifically, we evaluate all combinations of $\sigma_{\text{init}} \in \{0.01, 0.02, 0.04\}$ and $\sigma_{\text{refine}} \in \{0.01, 0.02, 0.04, 0.05\}$. Table I reports the safe-and-success rate for each combination.

The results show a clear stage-dependent effect of the robustness margin. For $\sigma_{\text{init}} = 0.01$ – 0.02 , increasing σ_{refine} from 0.01 to 0.02–0.04 substantially improves the safe-and-success rate. In contrast, $\sigma_{\text{init}} = 0.04$ performs poorly across all refinement margins, indicating that refinement cannot fully recover from an overly conservative initial barrier. Overall, the best performance is obtained with a relatively small σ_{init} and a moderately larger σ_{refine} . We therefore use $\sigma_{\text{init}} = 0.01$ and $\sigma_{\text{refine}} = 0.02$ in the final model.

These results suggest that σ primarily controls the conservativeness of the learned safety boundary. A large σ_{init} compounds the conservativeness of the behavior-policy labels and excessively shrinks the initial safe set, whereas refinement provides more accurate near-boundary labels allowing a larger σ_{refine} to improve safety without the same performance degradation observed during initial training. The poor recovery from $\sigma_{\text{init}} = 0.04$ further suggests that initial training establishes interaction-dependent structure that subsequent refinement can only modify locally. Overall, the method remains effective over a range of moderate robustness margins, while performance degrades primarily when the initial margin becomes overly conservative.

TABLE II: Computational efficiency and real-world motion smoothness. Simulation compute times are measured over 100 parallel environments. Isaac Lab processes environments in parallel, so the compute time for 100 environments is not 100 times that of a single environment.

Domain	Scenario	Compute Time (s) ↓	Mean Jerk (mm/s ³) ↓
Simulation (100 env)	4 Objects	550.10	–
	10 Objects	1556.31	–
	20 Objects	3017.80	–
	40 Objects	6285.85	–
Real Robot (1 rollout)	Table (low friction)	45.51	1.44
	Towel (high friction)	43.06	1.74

VI. CONCLUSION

In this work, we introduced Dense Contact Barrier Functions (DCBF), a scalable safety framework for manipulation in densely cluttered environments where contact with surrounding objects may be unavoidable. By redefining safety from collision avoidance to per-object stability, we introduce a data-driven safety estimate that is separate from the nominal task controller and can be paired with different nominal controllers, as it explicitly separates safety from task execution. Trained on scenes of only four objects, the learned barrier transfers seamlessly to 40 heterogeneous objects in simulation and zero-shot to hardware clutter of comparable density. While our results are promising, several exciting directions for future work remain. We plan to validate our method’s ability to handle more diverse objects, including those with varying observable parameters and additional sensing modalities, and to extend it to unconstrained full-body motion and deformable manipulation. This research marks a critical step toward deploying robots that can safely and competently maneuver through the unstructured, contact-rich environments of the real world.

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